Math 81: Abstract Algebra

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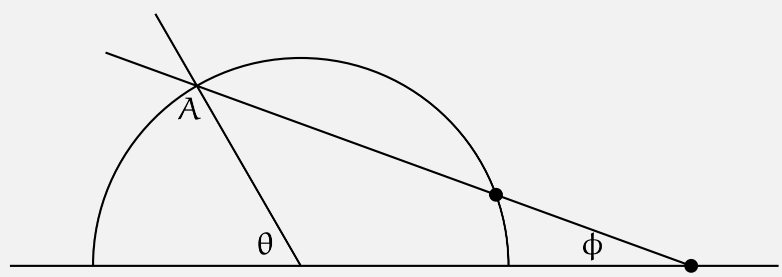
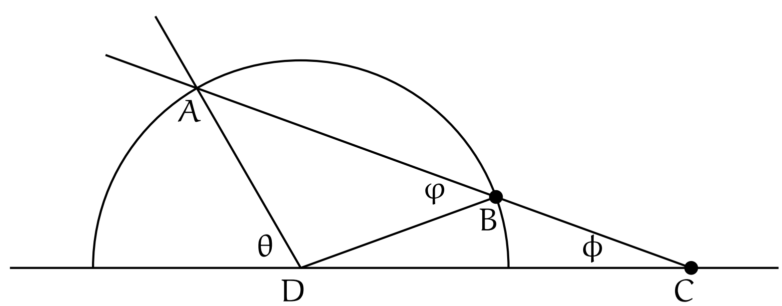
Problem 1. Archimedes discovered a construction that trisects any given angle using a compass and a "marked" straightedge. This straightedge has two special markings a distance 1 apart. The straightedge can be placed so that it passes through any already drawn point and such that both markings intersect other already drawn lines or circles.



In the picture, start with a horizontal line and another line meeting it with angle θ. Draw a circle of radius 1 around the intersection point of these two lines. The other line meets the circle at point A. Then place the marked straightedge so that it passes through the point A and such that the markings intersect the circle and the horizontal line. Prove that the angle ϕ that the straightedge makes with the horizontal line is equal to $\theta/3$.

Solution.



Note that $BC = 1$ (by construction), $BD = AD = 1$ (radius of the circle), hence the triangles ABD and BCD are isoceles. (We can construct the segment BD because we already have the two points.)

Consider the triangle BCD. We have the equality,

$$180 = 2\phi + (180 - \varphi) \implies 2\phi = \varphi$$

Now consider the triangle ABD. We have the equality,

$$180 = 2\varphi + ((180 - \theta) - \phi) \implies \theta + \phi = 2\varphi$$

Substituting and solving gives us,

$$\begin{aligned} \theta + \phi &= 2\varphi \\ \theta + \phi &= 4\phi \\ \theta &= 3\phi \end{aligned}$$

Hence, the angle ϕ that the straightedge makes with the horizontal line is equal to $\theta/3$.

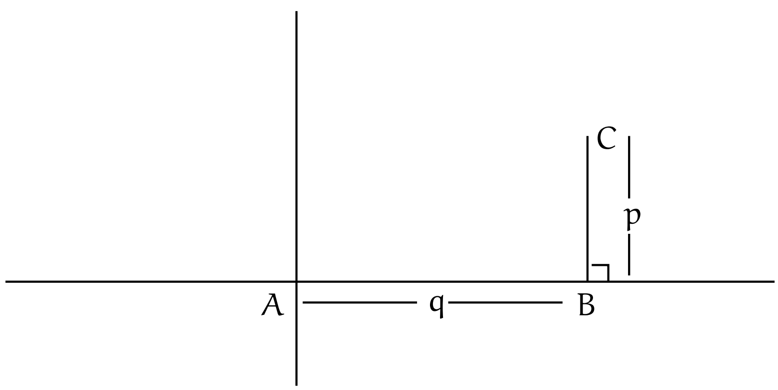
Problem 2. Prove that any angle θ such that $\tan(\theta)$ is rational can be constructed with compass and straightedge. Prove that such angles are dense in the interval $(-\pi/2, \pi/2)$.

Solution.

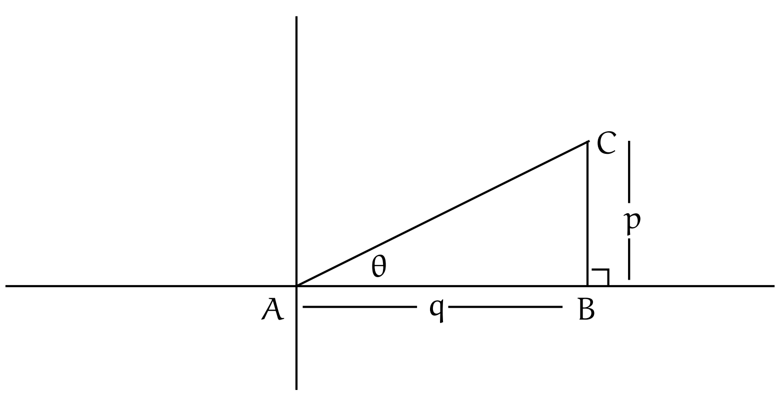
Let $\tan(\theta) = \frac{p}{q} \in \mathbb{Q}$, where $p, q \in \mathbb{Z}$. Then we can construct the angle θ by the following steps:

Note: We can construct line segments of length $|p|$ and $|q|$ because p, q are integers and hence in $\mathcal{P}_n$.

1. Draw a horizontal line segment of length $|q|$.
2. Construct a perpendicular to this segment at an endpoint of length $|p|$.



3. Connect points A and C. This gives us a right angled triangle with the angle θ between line segments AC and AB. (If $\tan(\theta)$ is negative, construct the perpendicular in the opposite direction.)



We know that the rationals are dense in $\mathbb{R}$, and $\tan^{-1} : \mathbb{R} \to [-\pi/2, \pi/2]$ is a bijective continous function. By defition, we know that for all points p in its domain for every $\epsilon > 0$, there exists a $\delta > 0$ such that $|x - p| < \delta$ implies $|f(x) - f(p)| < \epsilon$, i.e. dense sets are mapped to dense sets. That is, rational slopes (dense in $\mathbb{R}$) is mapped to constructibe angles (dense in $[-\pi/2, \pi/2]$).

Problem 3. Prove that an angle θ can be trisected using compass and straightedge if and only if the polynomial

$$4x^3 - 3x - \cos(\theta)$$

is reducible over $\mathbb{Q}(\cos(\theta))$.

Solution.

$(\implies)$

Assume an angle θ can be trisected using compass and straightedge. Let $\phi = \theta/3$. Then, from trignometry we know that the following equation holds,

$$\cos(\theta) = \cos(3\phi) = 4\cos^3(\phi) - 3\cos(\phi)$$

That is,

$$4\cos^3(\phi) - 3\cos(\phi) - \cos(\theta) = 0.$$

Since θ is constructible, so is $\cos(\theta)$, i.e. $\mathbb{Q}(\cos(\theta)) \subset \mathcal{P}_n$. From proposition 23 (D.F.), we know that if ϕ can be obtained from $\mathbb{Q}(\cos(\theta))$ using straightedge and compass constructions, then $[\mathbb{Q}(\cos(\theta))(\phi) : \mathbb{Q}(\cos(\theta)) = 2^k]$, for some k.

Since our polynomial is degree three, it must have at least one root in $\mathbb{Q}(\cos(\theta))$ and hence is reducible over that field.

$(\impliedby)$

Assume the polynomial $f(x) = 4x^3 - 3x - \cos(\theta)$ is reducible over $\mathbb{Q}(\cos(\theta))$, where θ is some constructable angle. Then there exists a root of $f(x)$, $\alpha \in \mathbb{Q}(\cos(\theta))$, i.e.,

$$f(x) = g(x)(x - \alpha)$$

Where $g(x)$ is some quadratic polynomial in $\mathbb{Q}(\cos(\theta))$. That is adjoining the roots of $g(x)$ to $\mathbb{Q}(\cos(\theta))$ is at most a degree two extension. Moreover, note that since θ is a constructable angle, so is $\cos(\theta)$. That is, $\mathbb{Q}(\cos(\theta)) \subset \mathcal{P}_n$. Because we know how to take square roots using compass and straightedge, any degree two extension of a field of constructable elements is constructable. Hence, the angle $\theta/3$ must be constructible (or θ can be trisected).

Problem 4. To 5-sect an angle means to divide it by 5.

1. Prove that the angles 2π, π, $2\pi/3$, and $\pi/2$ can be 5-sected by compass and straightedge.

2. Prove that a general angle cannot be 5-sected by compass and straightedge.

Solution.

1. (a) Note that, from Lemma 1.42 (F.T.), we know that the regular p-gon (p prime) is constructible if and only if $p = 2^n + 1$, $n \in \mathbb{Z}^+$. This holds for 5.

$$5 = 2^2 + 1$$

So, we can construct a 5-gon, with the central angle $2\pi/5$ That is, $2\pi/5$ is constructibe, and since we can bisect any angle, $\pi/5$ and $\pi/10$ are also constructable. Hence, we can 5-sect $2\pi, \pi, \pi/2$.

(b)

$$\frac{2\pi}{15} = \frac{5\pi}{15} - \frac{3\pi}{15} = \frac{\pi}{3} - \frac{\pi}{5}$$

Since we can construct both $\frac{\pi}{3}$, and $\frac{\pi}{5}$, we can subtract them and construct $\frac{2\pi}{15}$.

2. Let $\theta = \cos^{-1}(\frac{5}{7})$ We know that θ is constructable because $\cos(\theta) = \left(\frac{5}{7}\right)$ is constructable. Using the quintuple angle formula for cosine, we get

$$f(x) = 16y^5 - 20y^3 + 5y - \frac{5}{7} = 0$$

By Eisenstein, $f(x)$ is irreducible over $\mathbb{Q}(\cos(\theta)) = \mathbb{Q}$. Thus, the minimal polynomial for $\cos(\theta/5)$ has degree 5, and hence the extension $[\mathbb{Q}(\cos(\theta/5)) : \mathbb{Q}(\cos(\theta))] = 5$. As $\mathbb{Q}(\cos(\theta))$ is a constructible field and 5 is not a power of 2, $\mathbb{Q}(\cos(\theta/5))$ is not constructibe.

Problem 5. Let p be an odd prime.

1. Prove that both $\mathbb{Q}(\zeta_p)$ and $\mathbb{Q}(\zeta_{2p})$ have degree $p-1$ over $\mathbb{Q}$.
2. Prove that both $\mathbb{Q}(\cos(2\pi/p))$ and $\mathbb{Q}(\cos(\pi/p))$ have degree $(p-1)/2$ over $\mathbb{Q}$.
3. Find the minimal polynomial of $\cos(2\pi/9)$ over $\mathbb{Q}$ and prove that it splits completely over $\mathbb{Q}(\cos(2\pi/9))$.

Solution.

1. $\mathbb{Q}(\zeta_p)$

   ζ_p is a solution to $x^p-1=0 \implies x^p-1=(x-1)(x^p-1+\cdots+x+1)=0$. Since, $\zeta_p \neq 1$, ζ_p is a solution to $(x^p-1+\cdots+x+1)$. By Eisenstein, we know that $(x^{p-1}+\cdots+x+1)$ is irreducible. That is, the minimal polynomial of ζ_p is $(x^p-1+\cdots+x+1)$. Hence, the degree of the extension $[\mathbb{Q}(\zeta_p):\mathbb{Q}]=p-1$.

   $\mathbb{Q}(\zeta_{2p})$

   ζ_{2p} is a solution to $x^{2p}-1=0$, that is,

   $$x^{2p}-1=(x^p-1)(x+1)(x^{p-1}-x^{p-2}+\cdots-x+1)$$

   ζ_{2p} is not a solution to x^p-1 because $(\zeta_{2p})^p$ is not 1. And $\zeta_{2p} \neq -1$, then ζ_{2p} must be a solution to $(x^{p-1}-x^{p-2}+\cdots-x+1)$. By Eisenstein, we know that $(x^{p-1}-x^{p-2}+\cdots-x+1)$ is irreducible. That is, the minimal polynomial of ζ_{2p} is $(x^{p-1}-x^{p-2}+\cdots-x+1)$. Hence, the degree of the extension $[\mathbb{Q}(\zeta_{2p}):\mathbb{Q}]=p-1$.

   Therefore, both $\mathbb{Q}(\zeta_p)$ and $\mathbb{Q}(\zeta_{2p})$ have degree $p-1$ over $\mathbb{Q}$.

2. $\mathbb{Q}(\cos(\pi/p))$

   We know,

   $$\zeta_{2p}=e^{\pi i/p}=\cos(\pi/p)+i\sin(\pi/p)$$

   Consider the extension, $\mathbb{Q}(\cos(\pi/p))$. We know that cos is a real-valued function and hence, $\zeta_{2p} \notin \mathbb{Q}(\cos(\pi/p))$. Hence, $[\mathbb{Q}(\zeta_{2p}):\mathbb{Q}(\cos(\pi/2))] \geq 2$.

   Now consider the extension, $\mathbb{Q}(\cos(\pi/2))(i\sin(\pi/2))$. From the equation above we know that $\zeta_{2p} \in \mathbb{Q}(\cos(\pi/2))(i\sin(\pi/2))$. The following polynomial has $i\sin(\pi/2)$ as a root:

   $$f(x)=x^2-\cos^2(\pi/2)+1=0$$

   $i\sin(\pi/2)$ is a root to a degree 2 polynomial, hence the extension

$[\mathbb{Q}(\zeta_{2p}) : \mathbb{Q}(\cos(\pi/2))] = [\mathbb{Q}(\cos(\pi/2))(i\sin(\pi/2)) : \mathbb{Q}(\cos(\pi/2))] \leq 2$.
Since $[\mathbb{Q}(\zeta_{2p}) : \mathbb{Q}(\cos(\pi/2))] \geq 2$ and $[\mathbb{Q}(\zeta_{2p}) : \mathbb{Q}(\cos(\pi/2))] \leq 2$, then
$[\mathbb{Q}(\zeta_{2p}) : \mathbb{Q}(\cos(\pi/2))] = 2$.
Also from part(1), we know that $[\mathbb{Q}(\zeta_{2p}) : \mathbb{Q}] = p - 1$.

$$\begin{tikzcd} \mathbb{Q}(\zeta_{2p}) \\ \mathbb{Q}(\cos(\pi/p)) \arrow[u, "2"] \\ \mathbb{Q} \arrow[u, "n"] \arrow[uu, bend right=60, "p-1"'] \end{tikzcd}$$

Then, $n = (p-1)/2$ (by tower law).

$\mathbb{Q}(\cos(2\pi/p))$
Similarly,

$$\zeta_p = e^{2\pi i/p} = \cos(2\pi/p) + i\sin(2\pi/p)$$

That is, $\mathbb{Q}(\cos(2\pi/p)) \subset \mathbb{Q}(\zeta_p)$, with $\zeta_p \notin \mathbb{Q}(\cos(2\pi/p))$. And by the same logic the following two equations hold:

$$[\mathbb{Q}(\zeta_p) : \mathbb{Q}(\cos(2\pi/2))] \geq 2$$

and

$$[\mathbb{Q}(\zeta_p) : \mathbb{Q}(\cos(2\pi/2))] = [\mathbb{Q}(\cos(2\pi/2))(i\sin(2\pi/2)) : \mathbb{Q}(\cos(2\pi/2))] \leq 2$$

$\implies [\mathbb{Q}(\zeta_p) : \mathbb{Q}(\cos(2\pi/2))] = 2$

$$\begin{tikzcd} \mathbb{Q}(\zeta_p) \\ \mathbb{Q}(\cos(2\pi/p)) \arrow[u, "2"] \\ \mathbb{Q} \arrow[u, "n"] \arrow[uu, bend right=60, "p-1"'] \end{tikzcd}$$

$\implies n = (p-1)/2$ (by tower law).

3. The minimal polynomial of $\cos(2\pi/9)$ over $\mathbb{Q}$ is

$$f(x) = 4x^3 - 3x + \cos(2\pi/3) = 8x^3 - 6x + 1 = 0.$$

(Note $\cos(2\pi/9)$ is a root of $f(x)$, and $f(x)$ is irreducible by RRT, hence it must be minimal.)

Now consider the polynomial $f(x) \in \mathbb{Q}(\cos(2\pi/9))$. Since $\cos(2\pi/9)$ is a root of $f(x)$, the splitting field K of $f(x)$ is either a degree 1 or a degree two extension of $\mathbb{Q}(\cos(2\pi/9))$.

Note, that $\zeta_9 = \cos(2\pi/9) + i\sin(2\pi/9)$, and

$$\zeta_9^4 = \cos(8\pi/9) + i\sin(8\pi/9), \qquad \zeta_9^{-4} = \cos(8\pi/9) - i\sin(8\pi/9)$$

Then, $(\zeta_9^4 + \zeta_9^{-4})/2 = \cos(8\pi/9) \in \mathbb{Q}(\cos(2\pi/9))$.

We will show that $\cos(8\pi/9)$ is a root:

$$\begin{aligned}
f(\cos(8\pi/9)) &= 8((\zeta_9^4 + \zeta_9^{-4})/2)^3 - 6((\zeta_9^4 + \zeta_9^{-4})/2) + 1 \\
&= (\zeta_9^4 + \zeta_9^{-4})^3 - 3(\zeta_9^4 + \zeta_9^{-4}) + 1 \\
&= \zeta_9^{12} + \zeta_9^{-12} + 3(\zeta_9^4 - \zeta_9^4) - 3(\zeta_9^4 + \zeta_9^{-4}) + 1 \\
&= \zeta_9^3 + \zeta_9^{-3} + 1 \\
&= (\cos(2\pi/3) + i\sin(2\pi/3)) + (\cos(2\pi/3) - i\sin(2\pi/3)) + 1 \\
&= -\frac{1}{2} - \frac{1}{2} + 1 \\
&= 0
\end{aligned}$$

Hence, there exist at least two roots of $f(x) \in \mathbb{Q}(\cos(2\pi/9))$, and since $f(x)$ is a cubic, we are left with some linear polynomial and linear polynomials split over the field they are in. Hence, $f(x)$ completely splits over $\mathbb{Q}(\cos(2\pi/9))$.

Problem 6. For $3 \le n < 17$, determine whether a regular n-gon can be constructed by compass and straightedge.

Solution.

Again, from Lemma 1.42 (F.T), we know that a $p-$gon is constructable if and only if p is a Fermat prime.

1. $n = 3$

   3 is a Fermat prime, hence an equivalateral triangle is constructable.

2. $n = 4$

   Constructing a square is equivalent to constructing $\pi/2$, and we can do that by drawing a perpendicular segment at a point.

3. $n = 5$

   5 is a Fermat prime, hence a regular pentagon is constructable.

4. $n = 6$

   Inscribe an equilateral triangle in a circle. The point of intersection of the three perpendicular bisectors from a side to the opposite angle is the center of the circle, and the radius is half the length of any perpendicular bisector. Each side subtends a central angle of $2\pi/3$. Bisecting these angles produces six equal central angles $(\pi/3)$, whose endpoints form a regular hexagon.

5. $n = 7$

   7 is not a Fermat prime, hence a regular 7-gon is not constructable.

6. $n = 8$

   Inscribe a square in a circle. The point of intersection of the diagonals is the center of the circle, and the radius is half the length a diagonal. Each side subtends a central angle of $\pi/2$. Bisecting these angles produces eight equal central angles $(\pi/4)$, whose endpoints form a regular octagon.

7. $n = 9$

   To construct a regular $9-$gon, we want to trisect the central angle $(2\pi/3)$of an equivalateral triangle. To check if this is possible use Problem 3.

$$
\begin{aligned}
0 &= 4x^3 - 3x - \cos(2\pi/3) \\
0 &= 4x^3 - 3x + \frac{1}{2} \\
0 &= 8x^3 - 6x + 1 \\
0 &= (4x^2 - 1)(2x - 1)
\end{aligned}
$$

The equation is reducible over $\mathbb{Q}(\cos(\theta))$, hence, $2\pi/3$ is trisectable, and a regular $9-$gon is constructible.

8. $n = 10$

   Inscribe a regular pentagon in a circle. The point of intersection of the angle bisectors is the center of the circle, and the radius is the line segment between a vertice and the center of the circle. Each side subtends a central angle of $2\pi/5$. Bisecting these angles produces ten equal central angles ($\pi/5$), whose endpoints form a regular $10-$gon.

9. $n = 11$

   11 is not a Fermat prime, hence a regular 11-gon is not constructable.

10. $n = 12$

    Inscribe a regular hexagon in a circle. The point of intersection of the angle bisectors is the center of the circle, and the radius is the line segment between a vertice and the center of the circle. Each side subtends a central angle of $\pi/3$. Bisecting these angles produces ten equal central angles ($\pi/6$), whose endpoints form a regular $12-$gon.

11. $n = 13$

    13 is not a Fermat prime, hence a regular 13-gon is not constructable.

12. $n = 14$

    If a regular 14-gon were constructible, then the central angle $\pi/7$ would be constructible, and hence so would $2\pi/7$. This would imply the constructibility of a regular 7-gon, contradicting the lemma.

13. $n = 15$

    Inscribe an equilateral triangle in a circle. The point of intersection of the three perpendicular bisectors from a side to the opposite angle is the center of the circle, and the radius is half the length of any perpendicular bisector. Each side subtends a central angle of $2\pi/3$. From Problem 4 we know that $2\pi/3$ is 5-sectable. 5-secting these angles produces fifteen equal central angles ($2\pi/15$), whose endpoints form a regular 15-gon.

14. $n = 16$

Inscribe a regular octagon in a circle. The point of intersection of the angle bisectors is the center of the circle, and the radius is the line segment between a vertice and the center of the circle. Each side subtends a central angle of $\pi/4$. Bisecting these angles produces ten equal central angles ($\pi/8$), whose endpoints form a regular $16-$gon.